

CSD, Rennes

TD 1

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1 Problem 1 : State-Space Manipulation

Q1 Give the state-space representation, considering u and d as inputs, and y as the output.

We have a second order system, so the desired state-space is

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \star & \star \\ \star & \star \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \star \\ \star \end{bmatrix} u + \begin{bmatrix} \star \\ \star \end{bmatrix} d = Ax + Bu + B_d d,$$
$$y = \begin{bmatrix} \star & \star \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = Cx,$$

where the stars are to be filled.

There exist multiple possible choice. Let us choose $x_1 = y$, thus $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$. Let x_2 be the output of the first (the most left) block. Then $\dot{x}_1 = x_2 + d$.

Rewrite the transfer function as $\frac{100}{10+s}$. With the Observable Form, for x_2 we have

$$\dot{x}_2 = -10x_2 + 100u.$$

Finally,

$$\frac{d}{dt} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -10 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 100 \end{bmatrix} u + \begin{bmatrix} 1 \\ 0 \end{bmatrix} d,$$
$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Q2 Study the controllability with respect to u .

Compute

$$\mathcal{C} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 100 \\ 100 & -1000 \end{bmatrix}.$$

The matrix is of full rank, the system is completely controllable.

Q3 Define an augmented state which includes this variable as an internal variable. What is the state space representation with this augmented state space?

Define

$$x_e = \begin{bmatrix} x_1 \\ x_2 \\ d \end{bmatrix}.$$

As d is constant, it holds $\dot{d} = 0$. Then

$$\begin{aligned} \dot{x}_e &= \begin{bmatrix} 0 & 1 & 1 \\ 0 & -10 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ d \end{bmatrix} + \begin{bmatrix} 0 \\ 100 \\ 0 \end{bmatrix} u = A_e x_e + B_e u \\ y &= [1 \quad 0 \quad 0] x_e. \end{aligned}$$

Q4 Study its controllability, its observability

From the scheme, we can see that the new state d does not depend on u and we cannot control it. To check this intuition, compute

$$C_e = [B_e \quad A_e B_e \quad A_e^2 B_e] = \begin{bmatrix} 0 & 100 & -100 \\ 100 & -1000 & 10000 \\ 0 & 0 & 0 \end{bmatrix}.$$

The rank is 2.

For the observability, we compute

$$\mathcal{O}_e = \begin{bmatrix} C_e \\ C_e A_e \\ C_e A_e^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -10 & 0 \end{bmatrix}.$$

The matrix is of full rank, and the system is completely observable.

Q5 Proceed to the controllability canonical decomposition

The canonical decomposition is of the form

$$\begin{bmatrix} \dot{x}_c \\ \dot{x}_{uc} \end{bmatrix} = \begin{bmatrix} A_{cc} & A_{cu} \\ 0 & A_{uu} \end{bmatrix} \begin{bmatrix} x_c \\ x_{uc} \end{bmatrix} + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u,$$

where the dimension of x_c is the rank of the controllability matrix, which equals 2. Our system is already in this form by setting $x_{uc} = d$.

2 Problem 2 : Linearization model of a food network

$$\begin{cases} \dot{x}_1 = -x_1 + x_1 x_2 \\ \dot{x}_2 = -x_2 - x_1 x_2 + x_2 x_3 \\ \dot{x}_3 = -x_3 - x_2 x_3 + u \end{cases}$$

Q1 Given a constant food quantity u_0 , what are the different analytical expressions of the equilibrium points?

At any equilibrium it holds

$$\begin{cases} x_1 (x_2 - 1) = 0, \\ x_2 (x_3 - x_1 - 1) = 0, \\ x_3 (x_2 + 1) = u_0. \end{cases}$$

There are three possible equilibriums:

$$x_0^1 = \begin{bmatrix} 0 \\ 0 \\ u_0 \end{bmatrix}, x_0^2 = \begin{bmatrix} 0 \\ u_0 - 1 \\ 1 \end{bmatrix}, x_0^3 = \begin{bmatrix} \frac{u_0}{2} - 1 \\ 1 \\ \frac{u_0}{2} \end{bmatrix}.$$

Q2 Give the expression of the linearized models around with equilibrium points.

Let (x_0, u_0) be an equilibrium and define $\tilde{x} := x - x_0$ and $\tilde{u} := u - u_0$. Then

$$\dot{\tilde{x}} = \underbrace{f(x_0, u_0)}_0 + \underbrace{\frac{\partial f(x_0, u_0)}{\partial x}}_A \tilde{x} + \underbrace{\frac{\partial f(x_0, u_0)}{\partial u}}_B \tilde{u} = A(x_0, u_0) \tilde{x} + B(x_0, u_0) \tilde{u}.$$

Compute

$$A(x_0, u_0) = \begin{bmatrix} -1 + x_{0,2} & x_{0,1} & 0 \\ -x_{0,2} & -1 - x_{0,1} + x_{0,3} & x_{0,2} \\ 0 & -x_{0,3} & -x_{0,2} - 1 \end{bmatrix}, B(x_0, u_0) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Around the defined equilibrium points, it yields

$$A^1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & u_0 - 1 & 0 \\ 0 & -u_0 & -1 \end{bmatrix}, A^2 = \begin{bmatrix} u_0 - 2 & 0 & 0 \\ 1 - u_0 & 0 & u_0 - 1 \\ 0 & -1 & -u_0 \end{bmatrix}, A^3 = \begin{bmatrix} 0 & \frac{u_0}{2} - 1 & 0 \\ -1 & 0 & 1 \\ 0 & -\frac{u_0}{2} & -2 \end{bmatrix}.$$

Q3 We suppose now that $u_0 = 3$. Study the stability and the controllability of the linearized models.

For the first equilibrium, we have

$$A^1 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & -3 & -1 \end{bmatrix}.$$

We see that x_1 depends only on x_1 and it is stable. For x_2 it is inverse – the system is not stable. For info, in Matlab, the eigenvalues are $-1, -1, 2$. It is not stable. The controllability matrix is

$$C = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix}.$$

The rank is 1. The states x_1 and x_2 are not controllable.

For the second equilibrium, we have

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 0 & 2 \\ 0 & -1 & -3 \end{bmatrix}.$$

We see that x_1 depends only on x_1 and it is unstable – the system is not stable. For info, in Matlab, the eigenvalues are $-2, -1, 1$. It is not stable. The controllability matrix is

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & -6 \\ 1 & -3 & 7 \end{bmatrix}.$$

The rank is 2. The state x_1 is not controllable.

For the third equilibrium, we have

$$A^3 = \begin{bmatrix} 0 & \frac{1}{2} & 0 \\ -1 & 0 & 1 \\ 0 & -\frac{3}{2} & -2 \end{bmatrix}.$$

No intuition here, but Matlab says that the eigenvalues are $-1, -\frac{1}{2} - \frac{\sqrt{3}}{2}i, -\frac{1}{2} + \frac{\sqrt{3}}{2}i$. It is stable. The controllability matrix is

$$\mathcal{C} = \begin{bmatrix} 0 & 0 & \frac{1}{2} \\ 0 & 1 & -2 \\ 1 & -2 & -\frac{5}{2} \end{bmatrix}.$$

The rank is 3, it is controllable

Q4 What do you think on the ability to ensure that the top predator population remains constant?

Around the x_0^3 equilibrium point, x_1 stabilizes. However, it is not obvious what may happen for other starting domains.